

Aedes Mosquito Image Classification

Using ResNet50 Transfer Learning

Dataset:

Aedes Mosquito Image Dataset Version 1.0 (AMID v1)

Model:

ResNet50 — ImageNet Transfer Learning

Number of Classes: 4

Total Selected Images: 2000

Dataset Information

Dataset path (root): /content/aedes_dataset/AMID V1

Number of classes: 4

Class names: Aedes_aegypti, Aedes_albopictus, Culex_pipiens, Culex_quinquefasciatus

Images per class: 500

Total images: 2000

Train / Validation / Test split: 1400 / 300 / 300 (70% / 15% / 15%, stratified)

Model Architecture

Base: ResNet50, ImageNet pretrained weights, include_top=False

-> GlobalAveragePooling2D

-> Dropout(0.5)

-> Dense(256, activation='relu')

-> Dropout(0.3)

-> Dense(4, activation='softmax')

Trainable Parameters: 21,057,284

Non-trainable Parameters: 3,056,000

Total Parameters: 24,113,284

Training Configuration

Optimizer: Adam

Stage-1 learning rate: 0.0001

Fine-tuning learning rate (if used): 0.00001

Batch size: 32

Max epochs per stage: 30

Loss: categorical_crossentropy

Augmentations (train only): RandomRotation, RandomFlip, RandomZoom, RandomTranslation (width/height shift), RandomBrightness

EarlyStopping: monitor=val_loss, patience=7, restore_best_weights=True

ReduceLROnPlateau: monitor=val_loss, factor=0.2, patience=3, min_lr=1e-7

ModelCheckpoint: saves best model to results_ResNet50/best_model.keras

Fine-tuning used: True

Performance Summary

Model Name: ResNet50

Training Accuracy: 81.79%

Validation Accuracy: 77.67%

Testing Accuracy: 77.67%

Precision: 79.00%

Recall: 77.67%

F1 Score: 76.53%

Macro F1: 76.53%

Weighted F1: 76.53%

ROC AUC: 96.01%

Trainable Parameters: 21,057,284

Total Parameters: 24,113,284

Training Time: 22.64 min

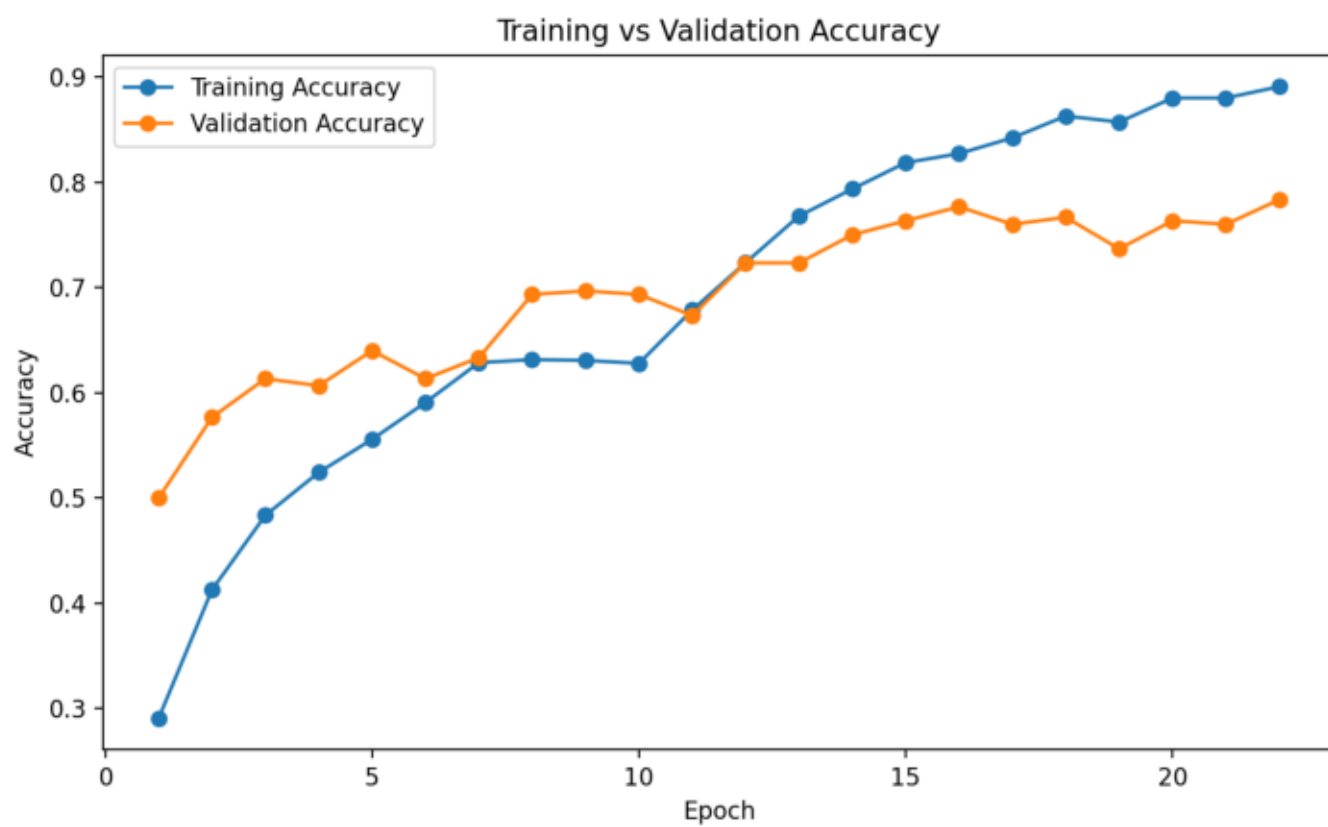
Inference Time: 4.24 ms/image

Fine-Tuning Used: True

Classification Report

	precision	recall	f1-score	support
Aedes_aegypti	0.6410	0.6667	0.6536	75
Aedes_albopictus	0.9048	0.5067	0.6496	75
Culex_pipiens	0.8353	0.9467	0.8875	75
Culex_quinquefasciatus	0.7789	0.9867	0.8706	75
accuracy			0.7767	300
macro avg	0.7900	0.7767	0.7653	300
weighted avg	0.7900	0.7767	0.7653	300

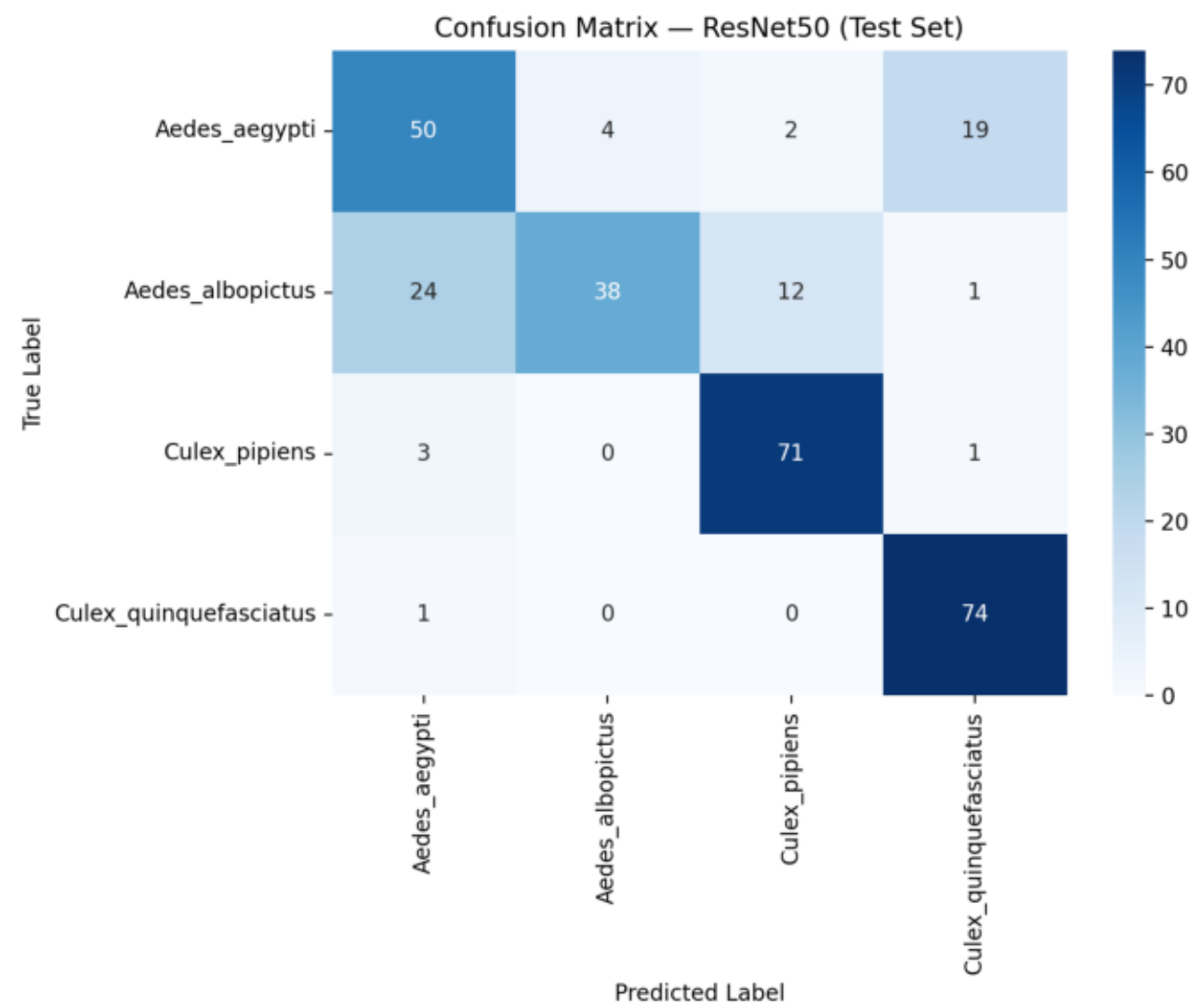
Accuracy Curve



Loss Curve



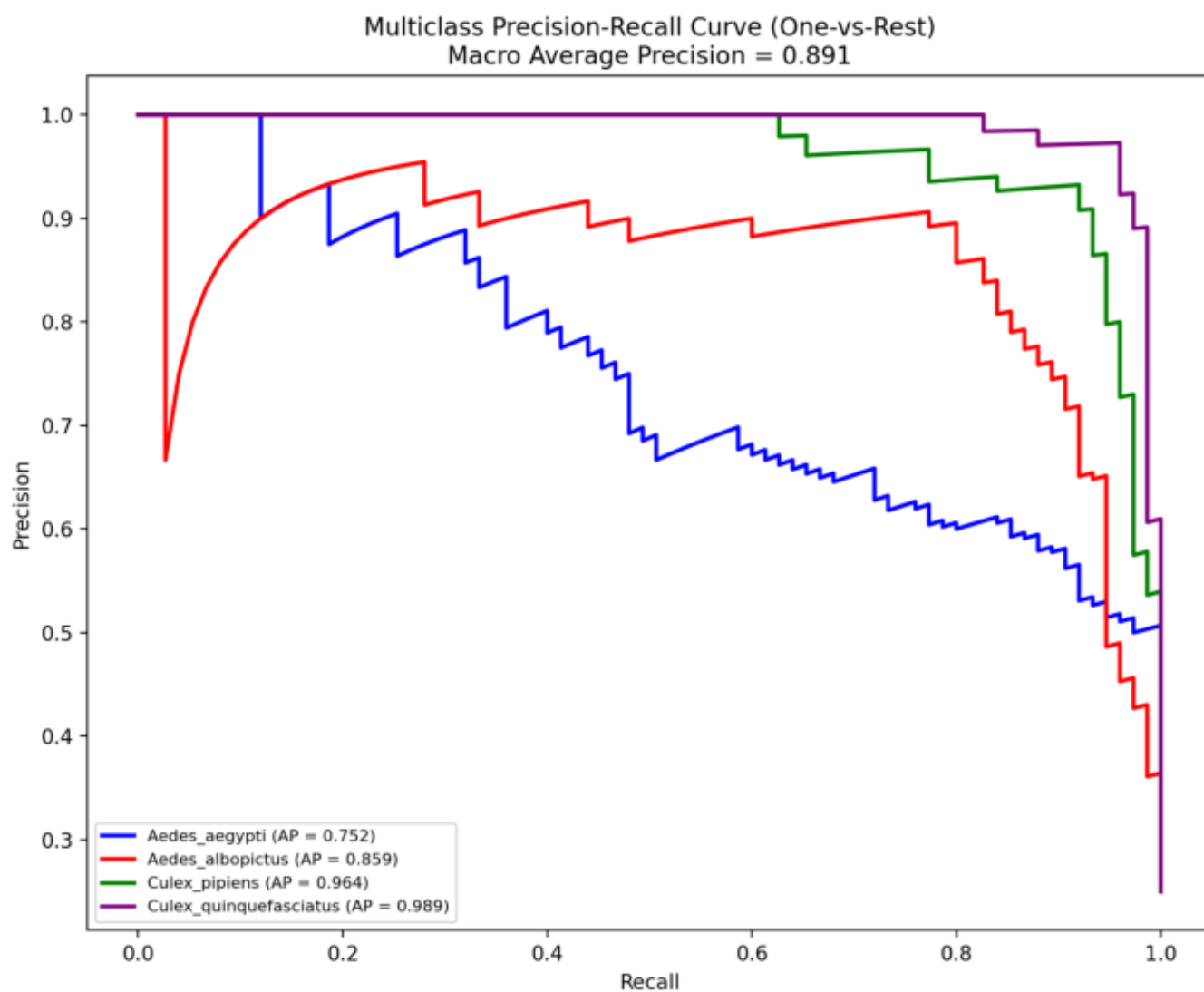
Confusion Matrix



Multiclass ROC Curve (One-vs-Rest)

Figure not found:
results_ResNet50/roc_curve.png

Multiclass Precision-Recall Curve (One-vs-Rest)



Sample Predictions

Figure not found:
results_ResNet50/sample_predictions.png

Final Results

Model Name: ResNet50

Training Accuracy: 81.79%

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